

# A nonseparable extended support for a Lipschitz functional

Candidate counterexample to a subquestion in arXiv:2009.07663

2026-06-21

## Status

This packet records a candidate counterexample, likely valid, to the separability part of Question 1 in Aliaga–Pernecká [1]. It does not disprove the weaker countable-chain-condition alternative in that question. Indeed, the support constructed below still satisfies ccc.

## Source question

The source paper is Ramon J. Aliaga and Eva Pernecká, “Integral representation and supports of functionals on Lipschitz spaces”, arXiv:2009.07663 [1]. In the concluding remarks, after proving that  $\mathcal{S}(\phi) \cap M$  is separable for every  $\phi \in \text{Lip}_0(M)^*$ , they ask:

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of the results obtained in the previous sections can be expressed for  $\text{Lip}(M)$  in a simplified form.

To end the paper, let us mention some problems and questions that we have dealt with throughout this work but were not able to answer.

For any  $m \in \mathcal{F}(M)$ , its support  $\text{supp}(m)$  is always a separable subset of  $M$ . Hence, by Corollary 3.7, its extended support  $\mathcal{S}(m) = \overline{\text{supp}(m)}^{\mathcal{U}}$  is a separable subset of  $M^{\mathcal{U}}$ . One may ask whether this is true for any functional  $\phi \in \text{Lip}_0(M)^*$ . A partial step was done in Corollary 3.8, where we show that  $\mathcal{S}(\phi) \cap M$  is always a separable metric space.

**Question 1.** For a complete metric space  $M$  and a functional  $\phi \in \text{Lip}_0(M)^*$ , is  $\mathcal{S}(\phi)$  a separable subset of  $M^{\mathcal{U}}$ ? Or does  $\mathcal{S}(\phi)$  at least satisfy the countable chain condition?

Our other question involves Proposition 4.4. The argument used in the proof leads through Bochner integration, which is why we place separability, resp. regularity, requirements necessary for Bochner integrability. Note that this would be the case also if we wanted to apply directly Proposition 4.3 combined with Lebesgue’s dominated convergence theorem, so that we would only need to consider sequences of functions instead of nets.

**Question 2.** Can we remove the hypothesis that  $M$  is separable or  $\mu$  is regular from Proposition 4.4? That is, if  $\mu$  is any Borel measure on  $M$  such that  $\int_M \rho d|\mu| < \infty$ , does it induce an element of  $\mathcal{F}(M)$ ?

Lastly, we have seen in Section 5.2 that the method for reducing the setting into bounded metric spaces presented in [1] was helpful for extending results concerning majorizability from functionals avoiding infinity to full generality. This was thanks to its order-preserving nature. It might be interesting to study also how the isomorphism defined in (5.6) affects extended supports (see [1, Proposition 7.4] for a

We answer the first half of Question 1 negatively:  $\mathcal{S}(\phi)$  need not be a separable subset of  $M^{\mathcal{U}}$ .

## Definitions used

For a pointed complete metric space  $M$ ,  $\text{Lip}_0(M)$  denotes the real Banach space of Lipschitz functions  $f : M \rightarrow \mathbb{R}$  with  $f(0) = 0$ , with norm the Lipschitz constant. Its canonical predual is the Lipschitz-free space  $\mathcal{F}(M)$ , and  $\text{Lip}_0(M)^*$  is the bidual  $\mathcal{F}(M)^{**}$ .

Aliaga–Pernecká define the extended support of  $\phi \in \text{Lip}_0(M)^*$  as

$$\mathcal{S}(\phi) = \bigcap \{K \subset M^{\mathcal{U}} : K \text{ compact and } \phi \in \mathcal{I}(K)^\perp\},$$

where  $M^{\mathcal{U}}$  is the uniform compactification and

$$\mathcal{I}(K) = \{f \in \text{Lip}_0(M) : f^{\mathcal{U}} = 0 \text{ on } K\}.$$

We use their Proposition 4.7 [1, Proposition 4.7]: if  $\phi \in \text{Lip}_0(M)^*$  is induced by an almost Radon measure  $\mu$  on  $M^{\mathcal{U}}$ , then

$$\mathcal{S}(\phi) = \text{supp}(\mu).$$

## Idea of the proof

Take a nonseparable compact space that nevertheless carries a full-support Radon probability measure: the Cantor cube  $K = \{0, 1\}^I$  for uncountable  $I$ , with its Bernoulli product measure. Let  $\Gamma$  be a dense subset of  $K$ , but forget the topology inherited from  $K$  and give  $\Gamma$  the bounded discrete metric. Then the uniform compactification of  $\Gamma$  is  $\beta\Gamma$ , because every bounded function on a uniformly discrete space is uniformly continuous.

The dense inclusion  $\Gamma \rightarrow K$  is uniformly continuous from the discrete metric and extends to a continuous surjection  $\pi : \beta\Gamma \rightarrow K$ . Lift the Bernoulli measure from  $K$  to a Radon probability  $\nu$  on  $\beta\Gamma$ . Its support cannot be separable: otherwise its continuous image under  $\pi$  would be a separable compact subset of  $K$ , but the image is all of  $K$ .

Finally let  $\phi$  be the functional on  $\text{Lip}_0(\Gamma)$  obtained by integrating Stone–Čech extensions against  $\nu$ . Proposition 4.7 of the source paper identifies  $\mathcal{S}(\phi)$  with  $\text{supp}(\nu)$ , so the extended support is nonseparable.

**Lemma 1.** *Let  $I$  be an uncountable set and  $K = \{0, 1\}^I$  with the product topology. Then  $K$  is nonseparable and carries a Radon probability measure with full support.*

*Proof.* Let  $\lambda = (\frac{1}{2}\delta_0 + \frac{1}{2}\delta_1)^I$  be the product probability measure. Every nonempty basic open cylinder has measure  $2^{-n}$  for some finite  $n$ , so  $\lambda$  has full support. Since  $K$  is compact Hausdorff,  $\lambda$  is Radon.

For the nonseparability, we use the standard density calculation. If  $D \subset 2^I$  is dense, then different coordinates must have different restrictions to  $D$ . Indeed, if coordinates  $i \neq j$  had  $x(i) = x(j)$  for every  $x \in D$ , then  $D$  would miss the nonempty basic open set  $\{x : x(i) = 0, x(j) = 1\}$ . Thus the map

$$I \longrightarrow 2^D, \quad i \longmapsto (x(i))_{x \in D}$$

is injective, so  $|I| \leq 2^{|D|}$ . In this packet we fix  $|I| > 2^{\aleph_0}$ , and then no countable  $D$  can be dense.  $\square$

**Lemma 2.** *Let  $\Gamma$  be a dense subset of the compact space  $K$ , equipped with the discrete metric  $d(\gamma, \eta) = 1$  for  $\gamma \neq \eta$ . If  $\lambda$  is a Radon probability measure on  $K$ , then there is a Radon probability measure  $\nu$  on  $\beta\Gamma$  such that  $\pi_*\nu = \lambda$ , where  $\pi : \beta\Gamma \rightarrow K$  is the continuous extension of the inclusion  $\Gamma \hookrightarrow K$ .*

*Proof.* The inclusion  $\Gamma \rightarrow K$  is uniformly continuous because  $\Gamma$  has the discrete metric. Hence it extends continuously to the Stone–Cech compactification  $\beta\Gamma$ . Its image is compact and contains the dense set  $\Gamma$ , so it is all of  $K$ .

Define a positive norm-one functional on the subspace  $\{g \circ \pi : g \in C(K)\} \subset C(\beta\Gamma)$  by

$$g \circ \pi \mapsto \int_K g \, d\lambda.$$

By the positive Hahn–Banach extension theorem, it extends to a positive norm-one functional on  $C(\beta\Gamma)$ . The Riesz representation theorem gives a Radon probability  $\nu$  on  $\beta\Gamma$ . For every  $g \in C(K)$ ,

$$\int_{\beta\Gamma} g \circ \pi \, d\nu = \int_K g \, d\lambda,$$

which is precisely  $\pi_*\nu = \lambda$ . □

**Theorem 1.** *There exist a bounded complete metric space  $M$  and a positive functional  $\phi \in \text{Lip}_0(M)^*$  such that  $\mathcal{S}(\phi)$  is not separable.*

*Proof.* Choose a set  $I$  with  $|I| > 2^{\aleph_0}$ , put  $K = 2^I$ , and let  $\lambda$  be the full-support Bernoulli product measure from Lemma 1. Let  $\Gamma \subset K$  be the set of finite-support points:

$$\Gamma = \{x \in 2^I : \{i : x(i) = 1\} \text{ is finite}\}.$$

This set is dense in  $K$ . Choose the zero function in  $\Gamma$  as base point and equip  $M = \Gamma$  with the discrete metric

$$d(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

Then  $M$  is bounded and complete. Since  $M$  is uniformly discrete, every bounded function on  $M$  is uniformly continuous, so its uniform compactification is  $M^{\mathcal{U}} = \beta M$ . Also  $\text{Lip}_0(M)$  consists exactly of the bounded real functions on  $M$  vanishing at the base point, with an equivalent norm.

Let  $\pi : \beta M \rightarrow K$  be the extension of the inclusion, and let  $\nu$  be a Radon probability on  $\beta M$  with  $\pi_*\nu = \lambda$ , supplied by Lemma 2. Define

$$\langle f, \phi \rangle = \int_{\beta M} f^{\mathcal{U}} \, d\nu, \quad f \in \text{Lip}_0(M).$$

This is a positive bounded functional on  $\text{Lip}_0(M)$ , induced by the Radon measure  $\nu$  on  $M^{\mathcal{U}}$ . By [1, Proposition 4.7],

$$\mathcal{S}(\phi) = \text{supp}(\nu).$$

We claim that  $\text{supp}(\nu)$  is nonseparable. Since  $\pi_*\nu = \lambda$  and  $\text{supp}(\lambda) = K$ , one has

$$K = \text{supp}(\lambda) \subseteq \pi(\text{supp}(\nu)).$$

The reverse inclusion is automatic because  $\pi(\text{supp}(\nu)) \subset K$ , hence  $\pi(\text{supp}(\nu)) = K$ . If  $\text{supp}(\nu)$  were separable, then its continuous image  $K$  would be separable, contradicting Lemma 1. Thus  $\mathcal{S}(\phi)$  is nonseparable. □

**Remark 1** (What remains open). *This construction does not refute the ccc alternative in Question 1. In fact,  $\mathcal{S}(\phi) = \text{supp}(\nu)$  has ccc: if  $\{U_i\}_{i \in A}$  is a pairwise disjoint family of nonempty relatively open subsets of  $\text{supp}(\nu)$ , then  $\nu(U_i) > 0$  for every  $i$ , and finite additivity over finite subfamilies implies that only countably many such  $U_i$ ’s can exist.*

## Verification and novelty check

The proof is purely structural. The local run indexes were searched for arXiv id 2009.07663, the title, “extended support”, “countable chain condition”, “Cantor cube”, and “nonseparable support”; no duplicate packet or attempt was found. Bounded web searches on 2026-06-21 for exact phrases from Question 1 and for the paper title found no later explicit solution of the separability subquestion.

Human review should focus on the measure-lifting step in Lemma 2 and on the invocation of Aliaga–Pernecká Proposition 4.7 identifying extended support with the support of the inducing Radon measure.

## References

- [1] R. J. Aliaga and E. Pernecká, *Integral representation and supports of functionals on Lipschitz spaces*, arXiv:2009.07663, 2020.